

The Political Economy of AI Infrastructure: Sustainability, Access, and Inequality in Large Language Models

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On my honor as a University Student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments

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Introduction

Over the past several years, large language models have been used for both research purposes and practical applications in education, programming, and writing. As these models become more advanced, they require increasing amounts of computing power, specialized hardware, and supporting infrastructure. In this sense, the development of artificial intelligence goes beyond the design of neural networks and machine learning algorithms; the availability of computing capacity, the cost of infrastructure, and the organizations that own this infrastructure also play an important role.

This raises a sociotechnical question of who has the ability to develop advanced AI models and under what infrastructural conditions. Currently, the infrastructure necessary to create and use large language models is primarily available to a limited number of large technology companies and cloud service providers. The rapid growth of AI infrastructure has also raised concerns about energy consumption, the need for cooling systems, and the overall sustainability of these developments.

This paper argues that the problems related to LLMs should be understood as issues concerning the infrastructure of artificial intelligence. First, access to high-end computing resources shapes who can participate in the development of these systems. Second, the operation of data centers and GPU-based systems raises sustainability concerns because of energy consumption, cooling needs, and natural resource use. Third, serverless architecture is one possible approach for rethinking how AI infrastructure is managed, though problems of sustainability and inequality would still remain.

Background and Context

As large language models continue to scale, their development has become increasingly dependent on large-scale computational infrastructure. This infrastructure includes GPUs, data centers, cloud platforms, storage systems, networking equipment, and cooling systems. These resources matter because developing advanced LLMs requires more than improving neural network design or machine learning algorithms. It also requires the ability to train, test and deploy models using expensive hardware over long periods of time.

This reliance on infrastructure affects different groups in different ways. Large technology companies can usually access GPU clusters, proprietary datasets, cloud platforms, and engineering teams that allow them to train and deploy large models. Academic researchers often work under more limited conditions. Instead of training frontier-scale models from scratch they may fine-tune open-source models, evaluate existing systems, or study smaller models because large-scale training requires computing resources that most universities cannot easily afford. This difference matters because infrastructure shapes not only who can build LLMs, but also what kinds of research questions can realistically be pursued.

Recent reports show this shift clearly. In 2023, industry produced 51 notable machine learning models, while academia produced 15 (Stanford Institute for Human-Centered Artificial Intelligence, 2024). This does not mean that academic researchers lack technical ability. Rather, it suggests that the ability to develop advanced AI systems increasingly depends on institutional access to computing infrastructure. Ahmed and Wahed (2020) describe this problem as the “compute divide,” in which participation in AI research and development depends on access to large-scale computational resources. Because of the high cost of infrastructure, smaller research groups and institutions may be limited in their ability to train large models, run repeated experiments, or contribute to cutting-edge systems.

The same infrastructure that creates access barriers also raises sustainability concerns. AI infrastructure requires electricity to power GPUs and servers, as well as cooling systems to keep data centers operating reliably. For example, training a single natural language processing model with extensive tuning can emit about 626,155 pounds of CO₂-equivalent, which shows the environmental cost associated with large-scale AI development (Strubell et al., 2020). More broadly, data centers that support large-scale AI systems consumed about 415 terawatt-hours of electricity globally in 2024 (IEA, 2024). These figures show that the environmental impact of AI is not limited to individual models. It is also connected to the larger infrastructure needed to train, deploy, and maintain them.

Taken together, these issues suggest that the challenges surrounding LLMs are not only about model performance. They are also about how AI infrastructure is built, accessed, and operated. The concentration of computing resources limits participation in AI development, while the operation of data centers creates long-term sustainability concerns. For this reason, researchers have begun to examine alternative computing models, including serverless architecture, that aim to reduce inefficient resource use and rethink how computing resources are allocated (Jonas et al., 2019).

Methods

For this research, a literature-based qualitative analysis is used as a research strategy to examine large language models as sociotechnical systems shaped by infrastructure, access, and sustainability. The purpose of this approach is to identify how LLM development depends on computational resources, how this dependence affects participation in AI research, and how alternative infrastructure models such as serverless architecture may address inefficient resource allocation.

The evidence is drawn from peer-reviewed journal articles, conference papers, preprints, and selected institutional reports focused on AI infrastructure, computational requirements, data center energy consumption, and serverless computing. Academic search strategies were used to identify relevant sources, using keywords such as “large language models compute cost,” “AI infrastructure sustainability,” “data center energy consumption,” “cloud infrastructure concentration,” and “serverless computing efficiency.” In addition to academic publications, institutional reports are included to provide empirical evidence on energy use and infrastructure trends.

Based on the research problem, the selected sources are evaluated in terms of how they contribute to understanding the relationship between AI systems and infrastructure. The evidence is organized around three themes that emerged from the literature: access to computational resources, sustainability of AI infrastructure, and serverless architecture as an alternative model for resource allocation. These themes are examined through a sociotechnical and political economy perspective, with attention to how technical systems, institutional resources, and infrastructure ownership shape the development and use of large language models.

This approach treats AI infrastructure as a sociotechnical system in which technical resources, institutions, and environmental conditions shape each other.

Analysis

This analysis examines LLM infrastructure through three connected issues: access to computing resources, sustainability of data center and GPU systems, and serverless architecture as an alternative model for resource allocation.

1. Access and the Compute Divide

A clear way to understand LLMs as infrastructure systems is through the compute divide. As Ahmed and Wahed (2020) describe, the compute divide is a situation in which participation in AI research becomes dependent on access to sufficient computational capacity. Technical skill and strong research ideas still matter, but access to hardware, funding, and institutional resources is also crucial for developing and training AI models, especially large language models.

This point becomes clearer when computation is considered in practical terms. For example, Le Scao et al. (2022) note that the training budget for the BLOOM project was set at 1,000,000 A100 GPU hours. One A100 GPU hour means using one NVIDIA A100 graphics processing unit for one hour. Therefore, one million GPU hours represents an immense amount of compute time for a large-scale project. This issue is important because it shows the cost of hardware and the difficulty smaller institutions face in accessing the equipment needed for large-scale model training.

Consequently, restricted access to computation affects what kinds of research can realistically be done. For example, a university lab may be able to fine-tune an existing model, test a chatbot, or evaluate a smaller model. However, it may not be able to train a frontier-scale model from scratch or run repeated training experiments to compare models of different sizes and settings. In this sense, limited compute means limited research capability and can shape the kinds of questions researchers are able to ask.

Access is also connected to cloud consolidation because cloud services play a critical role in providing compute resources. Many researchers depend on major cloud providers for GPU access, so pricing models, platform policies, switching costs, and egress fees can affect who is

able to use AI infrastructure. This means that access to AI development depends on technical ability as well as the economic structure of cloud infrastructure.

2. Sustainability and AI Infrastructure

The same infrastructure that creates access barriers also creates sustainability concerns. Large-scale AI systems require data centers, GPUs, servers, storage, networking equipment, and cooling systems. These systems consume electricity during training and deployment, and they also require energy and water to keep data centers operating reliably. For this reason, the sustainability of LLMs cannot be understood only by looking at individual models. It must also be understood at the infrastructure level.

Data center studies show that sustainability issues in AI involve more than total electricity use. Shehabi et al. (2024) analyze factors such as Power Usage Effectiveness and Water Usage Effectiveness. Power Usage Effectiveness, or PUE, measures how efficiently a data center uses electricity. Water Usage Effectiveness, or WUE, measures how much water a data center uses in relation to its computing activity. These metrics matter because a data center may improve energy efficiency while increasing water use for cooling. This shows that sustainability involves trade-offs across electricity, cooling, and local resource use.

The scale of this issue is significant. According to the International Energy Agency, data centers that support large-scale AI systems consumed about 415 terawatt-hours of electricity globally in 2024 (International Energy Agency, 2025). In the United States, data centers have also been estimated to consume around 66 billion liters of water directly for cooling in 2023, with hyperscale facilities accounting for most of this usage (Shehabi et al., 2024). These figures show that AI infrastructure can place pressure on both electricity systems and local water resources.

Hardware-level measurements also show why infrastructure details matter. Latif et al. (2024) report that a single 8xH100 HGX node running LLM training has a median power requirement of about 7.9 kW and a maximum of around 8.4 kW. This shows that even one high-performance training node can require substantial electricity. More importantly, it shows that AI power consumption depends on specific workloads, utilization rates, and infrastructure conditions. Without detailed infrastructure data, sustainability evaluations depend largely on what providers choose to disclose.

3. Serverless Architecture as an Alternative Infrastructure Model

Serverless architecture can be understood as one possible approach to allocating and managing AI infrastructure. Despite its name, a serverless model does not mean that there are no servers. Rather, it means that users do not directly manage fixed servers. The underlying infrastructure is managed by cloud providers, and computing resources are allocated when a task runs and released when the task is completed.

This approach is relevant for certain AI workloads where computing resources do not need to remain active permanently. Some tasks are occasional, meaning that they may not require a server or GPU to run all day. In these cases, serverless architecture may improve efficiency by reducing unnecessary resource allocation. This matters for sustainability because idle or underused infrastructure can still create energy use and cost.

Some studies suggest that serverless systems can improve performance and efficiency compared with traditional infrastructure in certain contexts. For example, Mankala and Silva (2025) found that serverless parallel processing on AWS reduced execution time by about 6.07 times, while also decreasing energy consumption by 81.2% and total costs by 79.1%. Akour and Alenezi (2025) also suggest that serverless computing may reduce energy use and costs, although

these benefits depend on the workload and the cloud provider. These findings show that the way infrastructure is organized can affect cost, energy use, and efficiency.

However, the advantages of serverless computing in some cases do not mean that this architecture solves the broader problems of AI infrastructure. Serverless computing still relies on cloud providers, which can create issues related to dependency, transparency, and control. It may also be less suitable for workloads that require continuous GPU memory availability, very low latency, or constant model availability. Therefore, serverless architecture should not be understood as a complete solution to inequality or sustainability. Its importance is that it shows AI infrastructure can be organized differently.

Conclusion

In conclusion, the challenges surrounding LLMs are, first and foremost, issues of infrastructure. The development of LLMs requires considerable computing power, which makes large technology companies major players in the development and improvement of AI models because they have access to GPU clusters, cloud platforms, and engineering resources. In contrast, smaller research teams are often limited to fine-tuning existing models, evaluating smaller systems, or studying models that are already available.

This infrastructure has two important consequences: unequal access and sustainability concerns. Data centers, GPU systems, cooling facilities, and cloud services require energy and water, which means that their operation creates significant sustainability challenges. For this reason, the environmental impact of LLMs should be discussed in terms of the underlying infrastructure rather than only specific models or individual training processes.

Serverless architecture offers one possible response to this problem. Serverless computing allocates computing resources when tasks are running and releases them once those tasks are completed, which may reduce unnecessary resource allocation in certain workloads. At the same time, serverless computing does not solve the broader issues of inequality, sustainability, or dependence on cloud providers.

From this perspective, this research demonstrates the importance of evaluating AI progress through infrastructure as well as model performance. Questions related to computing access, resource use, and infrastructure management should be considered when evaluating the development of LLMs. Future research could build on this work by comparing serverless computing with always-on infrastructure models.

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